

NGO Connect - Master Feed Architecture & AI Recommendation Roadmap

This roadmap is a long-term blueprint for building a world-class feed comparable in quality to Instagram, LinkedIn and Facebook, while optimizing for social impact, volunteer participation, NGO trust and meaningful engagement.

Phase 1 - Intelligent Rule-Based Feed

Candidate generation, weighted ranking, trust score, freshness, diversity, emergency override, caching, analytics, cursor pagination.

Phase 2 - Personalization

Learn user interests from behaviour, volunteer history, donations, searches and engagement.

Phase 3 - Recommendation Engine

Recommend NGOs, volunteers, campaigns and events using collaborative and content-based filtering.

Phase 4 - Machine Learning Ranking

Train ranking models using hundreds of features to predict engagement, volunteer conversions and donations.

Phase 5 - Embeddings

Use vector embeddings and semantic similarity for recommendations, search and matching.

Phase 6 - Real-Time AI

Streaming architecture, online inference, experimentation and continuous learning.

Core Modules

Feed API, Candidate Generator, Ranking Engine, Business Rules, Diversity Engine, Analytics, Cache, Admin Configuration.

Trust & Safety

Spam detection, fake NGO detection, moderation, report handling, trust scoring.

Technical Stack

ASP.NET Core (.NET), MySQL, Redis, Hangfire/Quartz, OpenSearch, Pinecone/pgvector, Docker, Nginx.

Future Deliverables

Detailed architecture, ERD, APIs, SQL strategy, sequence diagrams, ML roadmap, scalability guide and monitoring.